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| Course Code: MT-1004                             | Course Name: Linear Algebra |
| Instructor(s): Dr.Khusro Mian , Mr. Jamil Usmani |                             |
| Student Roll No:                                 | Section:                    |

**Instructions:**

- Return the question paper along with answer sheet. Scientific calculator is allowed.
- Read each question completely before answering it. There are 4 questions and 1 page.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.

**Time:** 60 minutes.

**Max Marks:** 30 points

**Question 1 [CLO-2] Estimated Time: 10 minutes Marks (5+5)**

- a) Consider the vectors  $v = (2, -1, 3)$ ,  $v_1 = (1, 0, 0)$ ,  $v_2 = (2, 2, 0)$ ,  $v_3 = (3, 3, 3)$

Find the coordinate vector of  $v$  relative to the basis  $S = \{v_1, v_2, v_3\}$  of  $R^3$

- b) Determine whether the polynomials are linearly dependent or independent in  $P_2$

$$P_1 = 1 - x, \quad P_2 = 5 + 3x - 2x^2, \quad P_3 = 1 + 3x - x^2$$

**Question 2 [CLO-2] Estimated Time: 25 minutes Marks (5+2+1)**

- a) Reduce  $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$  into row echelon form and find bases for null space and row space of A

- b) A matrix in reduced row echelon form is  $A = \begin{bmatrix} 1 & 0 & -2 & 0 & 1 \\ 0 & 1 & 3 & 0 & -4 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Find the Rank and Nullity of the matrix A

- c) Find the number of parameters in the general solution of homogenous linear system  $AX = 0$

for  $5 \times 7$  matrix of rank 3

**Question 3 [CLO-2] Estimated Time: 10 minutes Marks (4)**

Find the vector form of the general solution of  $AX = 0$  and find a bases and dimension for the solution space of homogeneous linear system

$$x_1 - 3x_2 + x_3 = 0$$

$$2x_1 - 6x_2 + 2x_3 = 0$$

$$3x_1 - 9x_2 + 3x_3 = 0$$

**Question 4 [CLO-2] Estimated Time: 15 minutes Marks (1+2+3+2)**

Find the characteristic equation, the eigenvalues and eigenvectors of the matrix  $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

and If A is diagonalizable, then find a matrix P that diagonalizes A and verify  $D = P^{-1}AP$